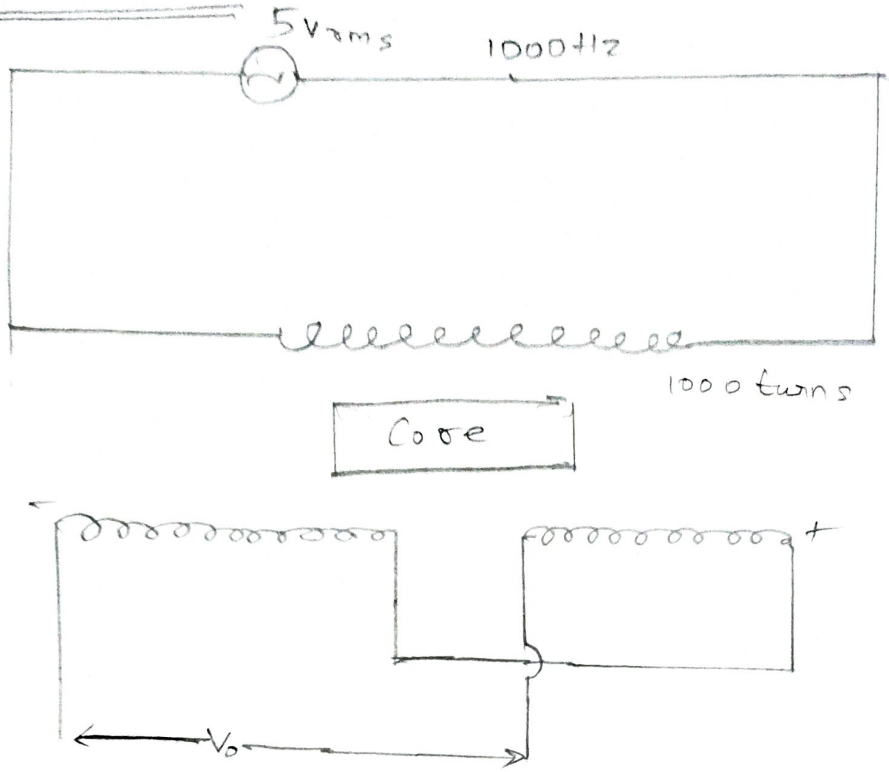
CIRCUIT DIAGRAM :



TABULATION :

	Input Voltage (V)	Output Voltage (V)	frequency (Hz	Time = $\frac{1}{f}$
Positive Displacement	7	2.2 V (D = 5mm)	1000	$\frac{1}{1000}$ = 1ms
Negative Displacement	7	(D = -5mm)	1000	1ms

CHARACTERISTICS OF LVDT (LINEAR VARIABLE DIFFERENTIAL TRANSFORMER)

AIM:

To understand and simulate the relation between core displacement and output voltage.

APPRATUS REQUIRED:

Laptop with internet connection.

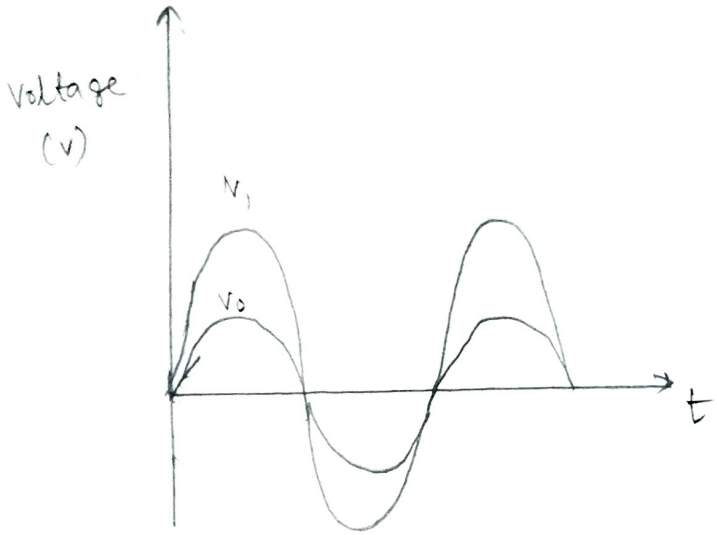
THEORY:

LVDT is an inductive transducer that converts linear displacement into a electrical signal. It consist of a transformer having one primary winding and two secondary windings wound on a core. As the primary is connected to an AC source, the AC current and the voltages are produced in the secondary of LVDT.

When the core is at the center, the flux liking with both the secondary winding are equal. So the emf induced in both the windings are equal, thus

MODEL GRAPH :

Positive Displacement.



Calculation :

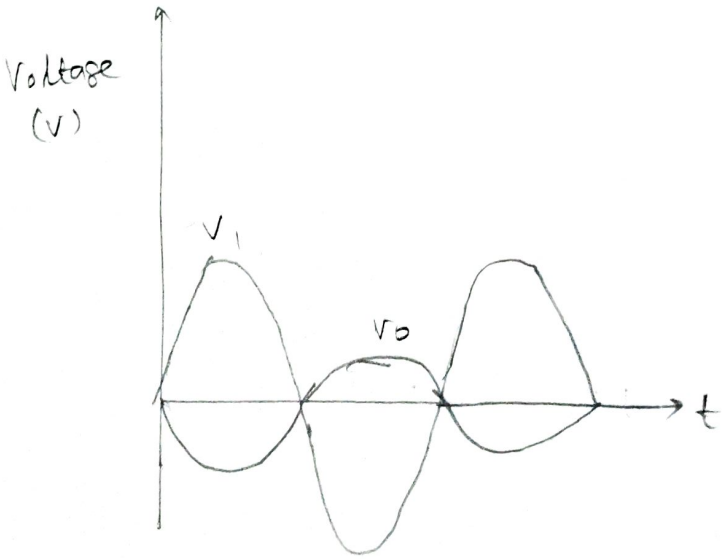
$V_m = \sqrt{2} \times V_{rms}$

$= \sqrt{2} \times 5$

$= 7\ V$

$T = 1ms$

Negative Displacement.



$4\ SD = 7$

$1\ SD = 7/4\ V$

$1.25\ SD = 7/4 \times 1.25$

$= 2.2\ V$

means there is no displacement. when the core is at the right, the flux linking with the secondary winding S_1 is more. So the emf induced in S_1 is more than S_2. So, the net emf is positive. When the core is at the left, the flux liking and the emf in S_2 is more than S_1. do, the net emf is negative.

PROCEDURE:

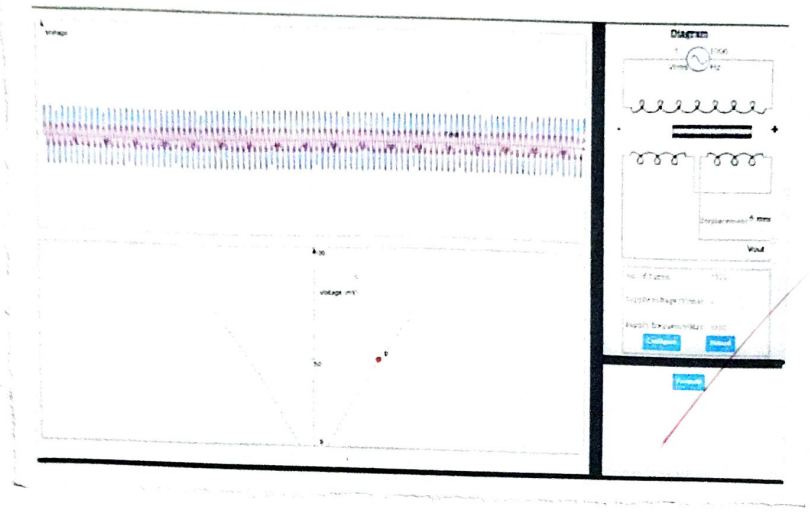
* Connections are made as per the circuit diagram.

* Set the number of turns, supplied voltage, and supplied frequency.

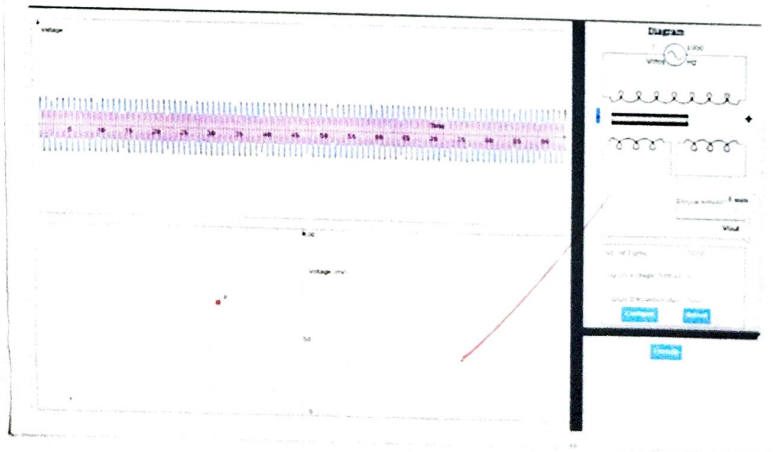
* Configure the parameters.

* Move the core to positive side and the negative side and observe the input voltage and output voltage waveforms

* Plot the graph between input and output voltage.



Positive Displacement



Negative Displacement

100

20/02/26

RESULT :

To understand and simulate the relation between core displacement and output voltage were developed and tested successfully.